

High Performance Scientific Computing: As with Big Data, Scientific Computing at a large scale is difficult to carry out using traditional machines. Weather and Climate forecasts, Natural disaster simulations, Computational model testing, etc., all depend on high performance computing (HPC) platforms to run. For HPC dedicated clusters are needed, something which not every institute in the world can afford. There is still a lot of research to be done in Cloud based HPC, as one of the problems is the varying internet bandwidth at the service or user end, which can cause solution errors. However, today it's possible to get supercomputing time from anywhere in the world for cheaper than ever before, all thanks to the cloud. For now, cloud computing for HPC is best for simulations without much inter-processor communication, but this is set to improve as well. Bioinformatics, Genome Sequencing, fluid flow runs are some of the HPC simulations that have been done on the cloud. Amazon EC2, Open Nebula and Nimbus are some examples of cloud based HPC services.



Defining cloud services

What exactly does a cloud service entail? Is it software, a computer machine, a mixture? A cloud service can be categorised broadly into the following:

Infrastructure as a Service (IaaS): As the name suggests the service includes provision of resources that support computing, storage and network and include machines and servers. Virtualisation allows the service to scale down or scale up as per the user's need. Services usually pool up these hardware resources from multiple servers and networks sourced from numerous data centers. The client is then given access to these pools to run their own platforms. Besides general hardware and computing infrastructure these services offer additional benefits such as raw storage, load balancers, firewalls, VLAN's and software bundles. There is utility-based costing and users pay only for the amount used or consumed. An IaaS typically can be accessed from anywhere and has no single node of failure. Examples of IaaS are Amazon EC2, NaviSite, Rackspace, Google Compute Engine and HP Cloud

Hardware as a Service (HaaS): This cloud computing service provision entails the provision of managed hardware that is leased for computing